Scientific Executive Report

Project: AI-driven demand forecasting and spatial optimization for Hokuriku tourism (Fukui Prefecture, Japan)\[3pt] **Author:** Amil Khanzada *Specially Appointed Associate Professor, Headquarters for Regional Revitalization, University of Fukui* **Date:** March 26, 2026

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1. Problem / 2. Data Architecture (DHDE)

1. Problem: "47th Place" and Economic Loss

Fukui Prefecture remains structurally weak in winter tourism (**47th/47**). Root cause is defined not as demand shortage but as "**Planning Friction**"—a gap between high digital intent and low physical visits, driven by weather uncertainty and lack of vibrancy, creating an Opportunity Gap.

2. Distributed Human Data Engine (DHDE)

Four data streams integrated: **Digital Intent** (Google search/route queries), **Environmental Filter** (JMA weather: temperature, precipitation, snow, wind), **Observed Data** (AI camera visitor counts), **Behavioral Sensor** (Hokuriku survey: 97,719 responses + 90,317 spending records).

3. Key Results (Forecast Accuracy & Kansei Threshold)

3.1 Forecast Performance & Weather Shield Effect

Accuracy: $R^2 = 0.810$ (adj. 0.802). 81% of daily visitor variation explained. Top predictor: Google "Directions" intent ($\beta = +0.456$). Adding JMA weather data boosts accuracy by +5.6%, proving weather as an economic gatekeeper.

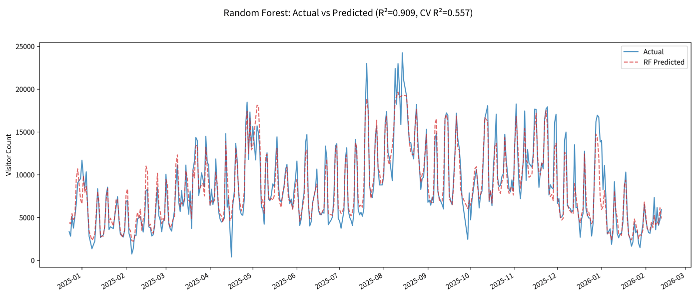

Figure 1: Demand forecast (red) vs AI camera actual (blue). $R^2 = 0.810$.

3.2 Under-vibrancy Paradox & Sacred Site Threshold

Text mining (71,623 reviews) reveals Fukui's essence is "under-vibrancy." Low satisfaction (1–2⋆) complaints about "loneliness/closed shops" are 11.5× more frequent. Tojinbo (nature) satisfaction rises with crowding; Eiheiji (sacred site) requires density management (threshold ≈ 42.4%).

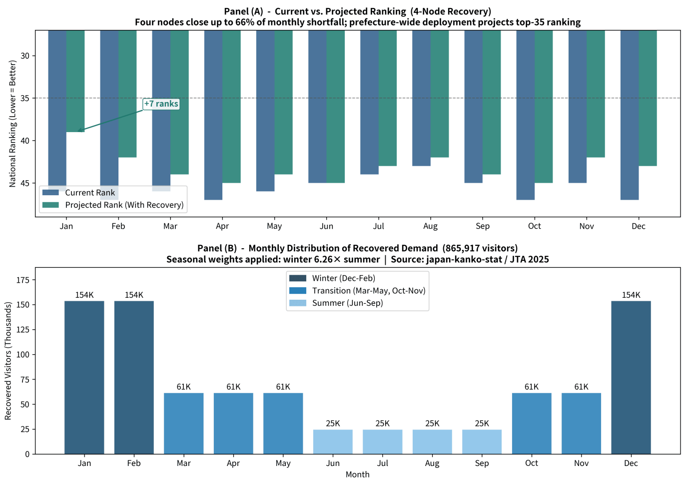

Figure 2: AI governance recovers 865,917 lost visitors, improving rank from 47th to ~35th.

4. Economic Impact & Regional Linkage

4.1 Opportunity Loss: ~¥11.96B (4 Nodes)

4 nodes (Tojinbo/North, Fukui Stn/Central, Katsuyama/East, Rainbow Line/South) achieved geographic saturation. Lost visitors: **865,917/year**. Estimated aggregate revenue loss: **~¥11.96B**. Winter sensitivity: **6.26**× higher than summer.

4.2 Ishikawa Pipeline (Regional Linkage Evidence)

Ishikawa tourism activity strongly leads Fukui visits ($r = 0.549$). Hokuriku functions as a single ecosystem—regional governance and joint grants are essential.

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5. Policy Proposals / 6. Conclusion

Policy (Recovering ~¥11.96B in lost demand): **(1) Supply-side Nudge** (Shop Activation Alert): Optimize opening hours/staffing 72 hours ahead based on demand forecast. **(2) Demand-side Nudge** (Weather Routing): Guide visitors from Tojinbo to indoor sites (Katsuyama, Eiheiji) during bad weather.

Four-Node Weather Shield Governance Architecture
Demand-Side Meteorological Rerouting Network, Fukui Prefecture, Japan

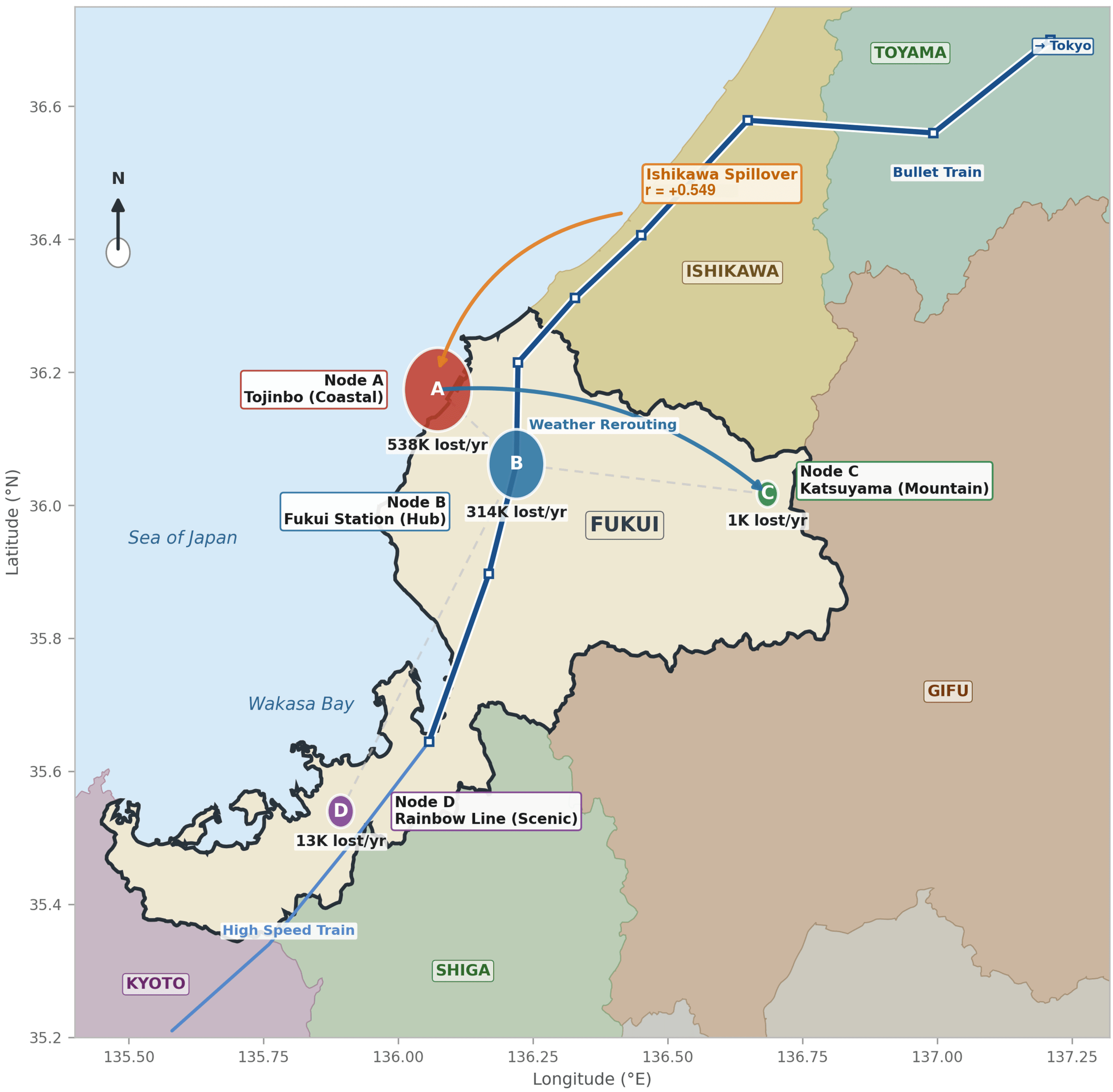

Figure 3: 4-node weather shield network. Geography-accurate map with weather sensitivity coefficients at each node. Rainbow Line shows strongest seasonality (1.85×) and snow impact ($\beta = -0.0916$).

Conclusion: DHDE achieves **full geographic saturation** (north, central, south, east). Connecting forecasts to AI nudges can recover ~¥11.96B in demand, raising Fukui's tourism economy from **47th to ~35th place**.